

Evaluation Report

Armenian Legal RAG Assistant - Law of the Republic of Armenia on Electronic Communications

1. Evaluation Objective

The benchmark compares three LLM providers while holding the retrieval evidence constant. For each question, the RAG system performs retrieval once, assembles one context, and sends that same context to OpenAI, Groq, and Gemini. This reduces retrieval variation and makes the comparison more representative of generation behavior and provider reliability.

The evaluation dataset contains 15 questions: 5 answerable Armenian questions, 5 answerable English questions, 3 out-of-scope/adversarial questions, and 2 synthesis questions.

2. Benchmark Results

Metric	OpenAI	Groq	Gemini
Successful requests	15 / 15	15 / 15	12 / 15
Answer accuracy	100.0%	100.0%	80.0%
Citation precision	96.7%	100.0%	70.0%
Citation recall	100.0%	100.0%	73.3%
Citation F1	97.8%	100.0%	71.1%
Hallucination proxy	0.0%	0.0%	0.0%
Failure rate	0.0%	0.0%	20.0%
Avg total latency	3,356 ms	5,706 ms	4,302 ms
Avg TTFT	N/A	5,704 ms	N/A
Avg prompt tokens	1,973.7	N/A	3,204.5
Avg completion tokens	92.4	N/A	88.6
Avg measured cost/request*	\$0.000937	N/A	\$0.002736

*Cost is calculated only where token usage was exposed by the provider integration. The benchmark used a paid-rate pricing snapshot even where actual usage was covered by credits or a free tier. Groq cost is reported as unavailable rather than estimated because token usage was unavailable in the current streaming integration.

3. Interpretation

OpenAI: Completed all 15 requests and achieved 100% automated answer accuracy with 97.8% citation F1. It also had the lowest measured average total latency among the three integrations in this benchmark. TTFT is unavailable because the current structured-output adapter is non-streaming.

Groq: Completed all 15 requests and achieved 100% automated answer accuracy and 100% citation F1. Its measured average total latency was 5.706 seconds. The observed TTFT was nearly equal to total latency because the first meaningful streamed content arrived near completion in this integration; this should not be generalized as an intrinsic property of the provider. Token usage was unavailable, so measured cost could not be calculated.

Gemini: Completed 12 of 15 requests. Three requests failed because of service unavailability and free-tier quota/rate-limit conditions, producing a 20% failure rate. Because failed requests remain in the end-to-end accuracy denominator, the resulting 80% accuracy does not by itself show that Gemini's successful answers were worse than those of the other providers.

4. Recommendation

For this implementation, OpenAI is the most practical default provider based on the observed combination of zero benchmark failures, strong citation performance, available token accounting, and the lowest measured average total

latency. Groq is a useful alternative based on its strong automated answer and citation results, but the current integration lacks token accounting and showed higher measured latency. Gemini should remain supported, but the observed free-tier availability and quota failures make it less reliable for this specific benchmark environment.

This recommendation is specific to this small benchmark, provider configurations, API tiers, and the time of measurement. It should be re-evaluated with a larger dataset and production-tier limits before a production deployment decision.

5. Metric Methodology

Answer accuracy. For answerable questions, generated and expected answers are compared using multilingual embedding similarity with a threshold of 0.75. For known unanswerable questions, correctness is based on the structured *insufficient_context* field. API failures count as incorrect in end-to-end accuracy.

Citation accuracy. Generated article numbers are compared with expected articles using precision, recall, and F1. Citations are also deterministically checked against the articles actually supplied in retrieved context.

Hallucination proxy. The automated proxy flags invalid citations and failure to abstain on known out-of-scope questions. A 0% proxy score does not prove the absence of unsupported claims inside otherwise valid answers.

Latency and failures. Total request latency is measured for successful provider calls. TTFT is recorded where a usable streaming path exists. Rate limits, service-unavailable responses, timeouts, and malformed outputs are retained as failures rather than discarded.

6. Important Limitations

- The dataset contains only 15 questions, so the results should not be treated as statistically broad model rankings.
- Automated semantic similarity is not equivalent to human legal review. A semantically similar answer can still omit or distort an important legal detail.
- The hallucination metric is only a proxy and does not perform claim-level entailment against retrieved evidence.
- Retrieval quality is not independently measured with Recall@K, MRR, or a larger labeled retrieval dataset.
- The benchmark was affected by free-tier provider limits, especially for Gemini.
- TTFT is unavailable for the current non-streaming OpenAI and Gemini structured-output integrations.
- Groq token counts were unavailable in the current streaming integration, preventing measured cost calculation.
- Latency measurements depend on network conditions, provider load, model configuration, and the small sample size.

7. Qualitative Legal-Answer Limitation

A qualitative Armenian test exposed an important limitation of automated scoring. The system retrieved and cited Article 2 for a definition question, but the generated wording was not a fully faithful reproduction of the legal definition. This demonstrates why high embedding similarity and a valid citation should not be presented as proof of legal correctness. Production use would require stronger groundedness checks and human legal review.

8. What I Would Improve With More Time

- Expand the benchmark and have legal-domain reviewers validate expected answers and generated answers.
- Evaluate retrieval separately using Recall@K and MRR.
- Introduce clause-aware legal chunking and improved Armenian lexical normalization.
- Add a reranking stage for difficult multi-article questions.
- Add claim-level groundedness or entailment evaluation.
- Persist the document and vector index instead of rebuilding embeddings at application startup.
- Add provider retries, fallback policies, circuit breaking, and production observability.

- Implement consistent streaming instrumentation so TTFT and token accounting are available across providers.
- Re-run cost analysis using a versioned pricing snapshot and production API tiers.

9. Conclusion

The benchmark demonstrates a working multilingual RAG evaluation pipeline in which retrieval is held constant across providers and operational failures are included in the results. The results are useful for selecting a default model for this prototype, but they should be interpreted as evidence from a small engineering benchmark rather than as a general model leaderboard or a substitute for legal-quality evaluation.